Analysis of Opportunities for International Students in Japan

Cloud Technologies and Big Data Frameworks

Politecnico di Milano - Graduate School of Management

Group 5:
Maxime Engels
Lucia Francescoli
Tiziano Gallo
Frank Novoa

Index

Datasets

Problem & Research Question

Queries

Conclusions

Datasets

Japan Top Universities Dataset

- 52 leading universities in Japan
- Research impact score
- International student ratio
- Graduate employment rate
- Institution type and region code

Japan Region Dataset

- 47 Japanese prefectures
- Population data (2024)

Japan Expenditure Dataset

- Average living expenditure in 47 japanese prefectures in Yen (2007)
- Regions and Region Code

Problem & Research Question

Problem Description

The Japanese higher education system is highly competitive and internationally recognized. Students who wish to study in Japan face complex decisions, considering factors such as internationalization, research impact, graduate employability, and regional living and education expenditures, which significantly influence the overall affordability of studying at different universities.

Research Question

Which universities and regions in Japan offer the best opportunities for international students, considering internationalization, research impact, employability, and regional living expenditure, and to detect geographical gaps or inequalities that could hinder the attraction of both global and local talent?

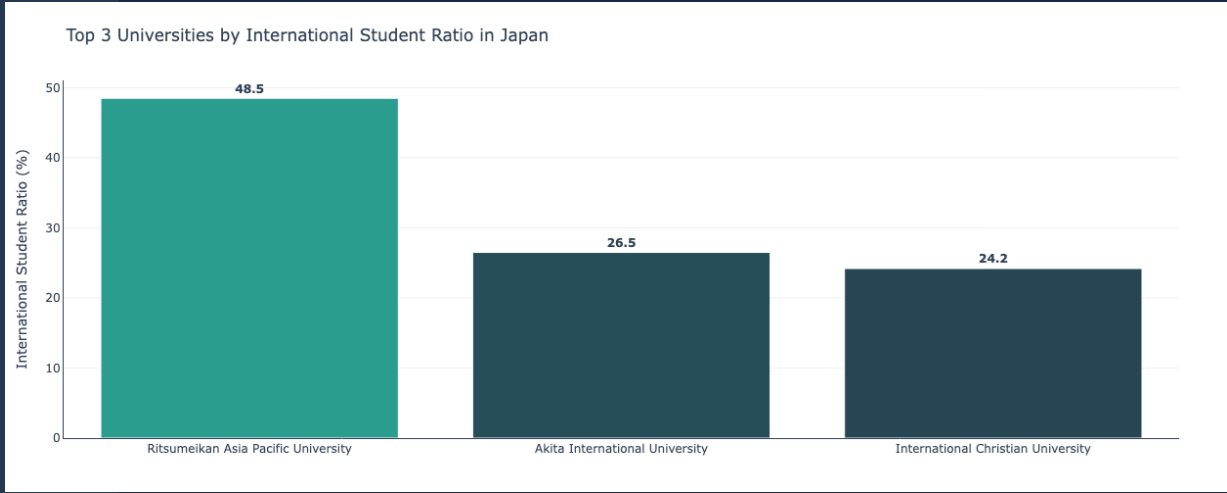
Queries

?

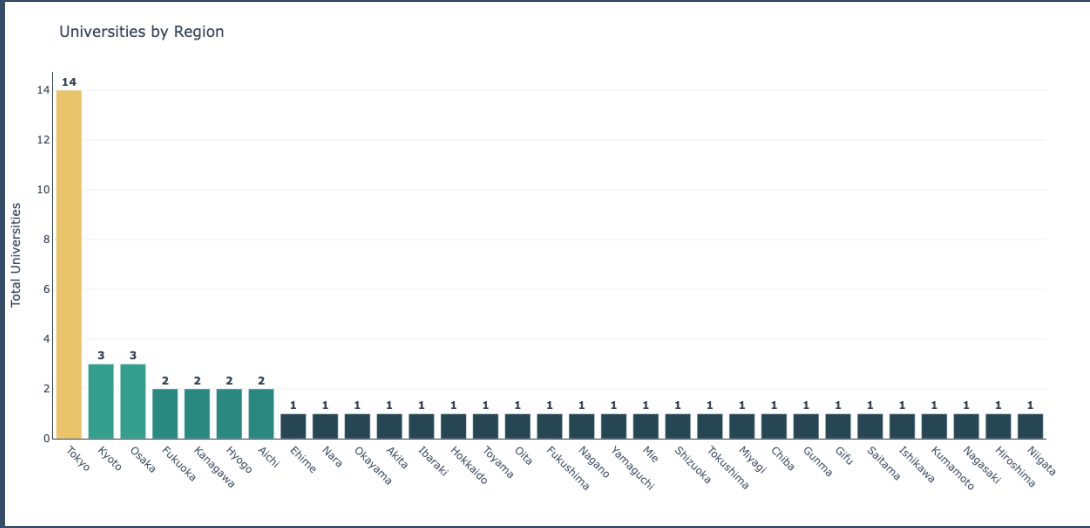
Highest International Student Ratios

→ Ritsumeikan Asia Pacific University shows the highest international student ratio.

→ Followed by Akita International University and International Christian University.



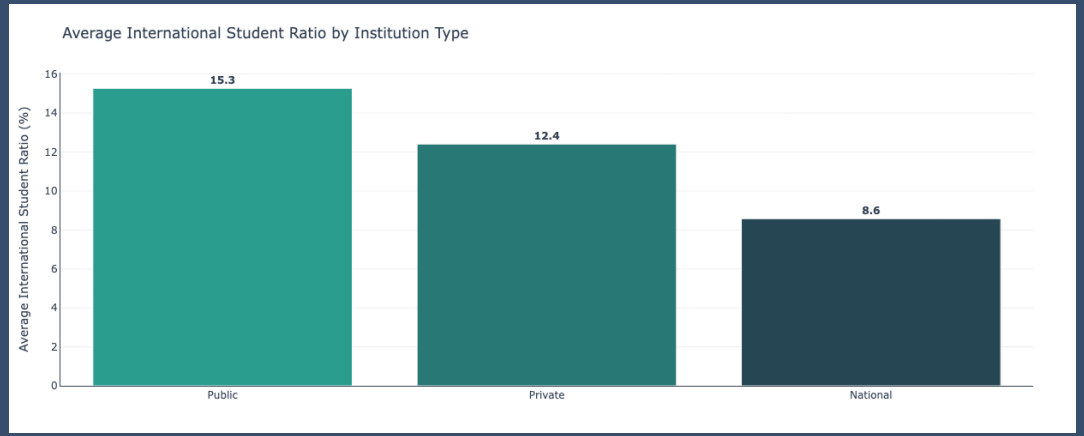
Region with Highest Number of Universities



→ Tokyo's concentration of 14 top universities offers unmatched program variety, study environments and opportunities for international students.

→ With Kyoto and Osaka both on 3 and Fukuoka, Kanagawa, Hyogo, and Aichi on 2 each.

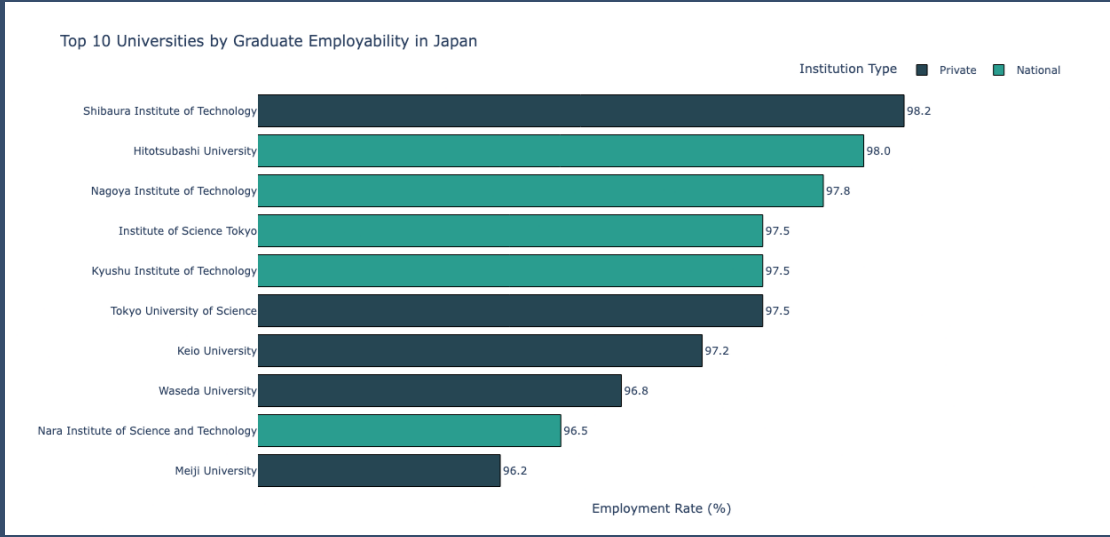
How does the average ratio of international students differ between institution types?



→ Public universities have a higher average proportion of international students compared to private and national universities.

→ This could suggests that public universities are relatively more open or attractive to international students.

Top 10 Universities by Graduate Employability

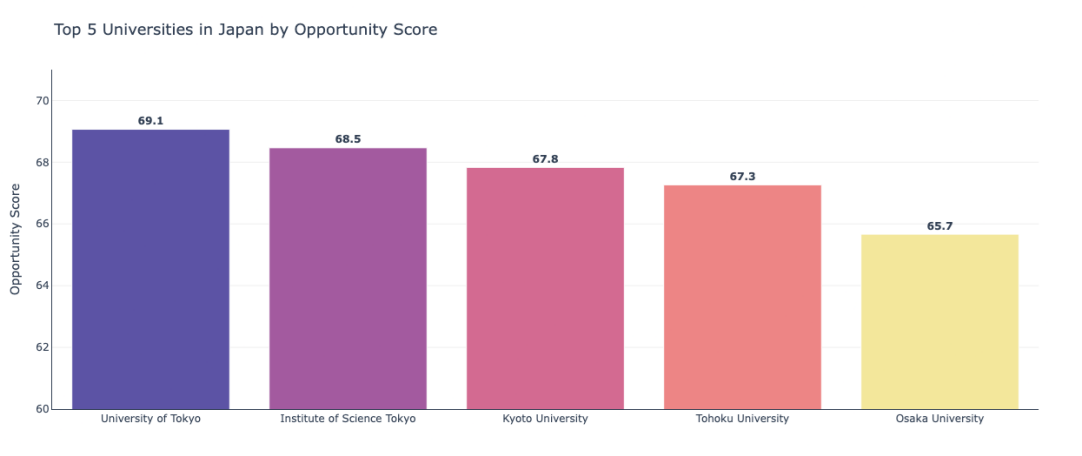


→ Private and National universities dominate employability rankings with strong industry connections.

→ Most Universities are mainly involved with Science and Technology.

→ Public universities do not appear in the top 10.

Top 5 Universities by Overall Opportunity Score

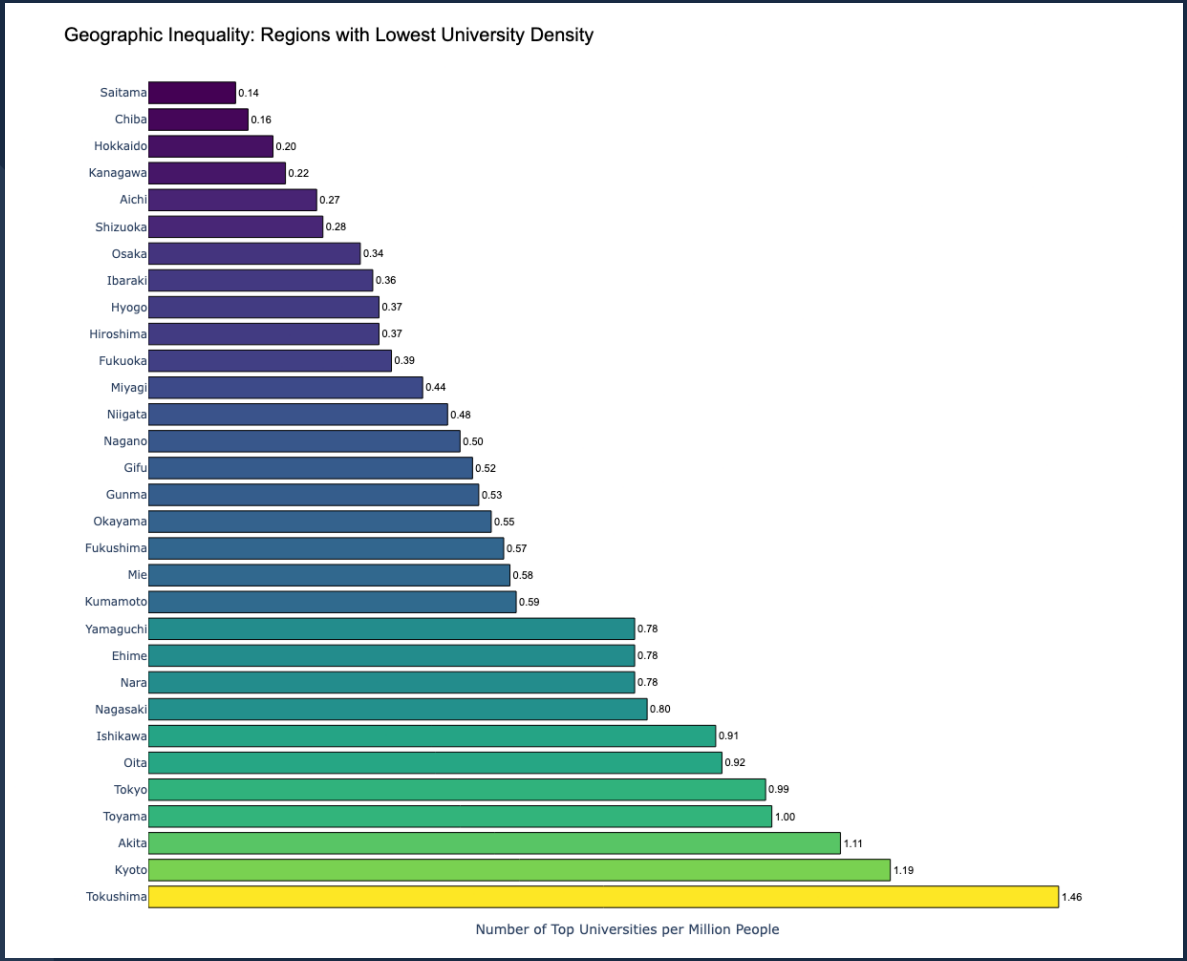


#1 University of Tokyo

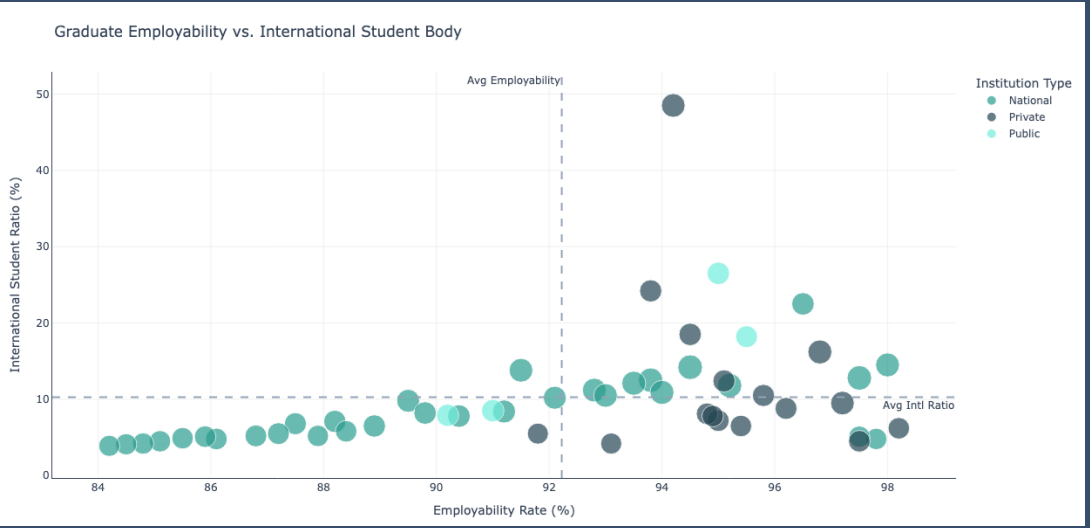
→ The university of Tokyo offers the best overall opportunities for international students seeking quality education, career prospects and an international environment.

Geographic Inequality

→ Saitama is the region with the most significant geographic inequality, having the lowest university density relative to its population.



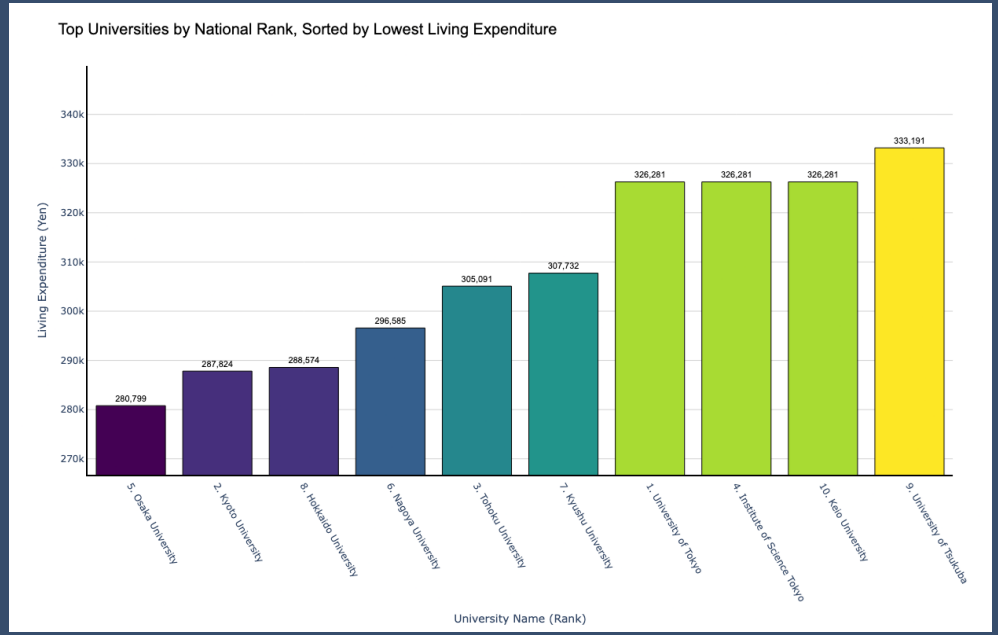
Employability Relation to Number of International Students



→ Universities with fewer international students usually have lower employment rates.

→ National universities seem to be top choices because they have both large numbers of international students and great job and career prospects.

Top 10 Universities: Sort by Lowest Living Expenditure



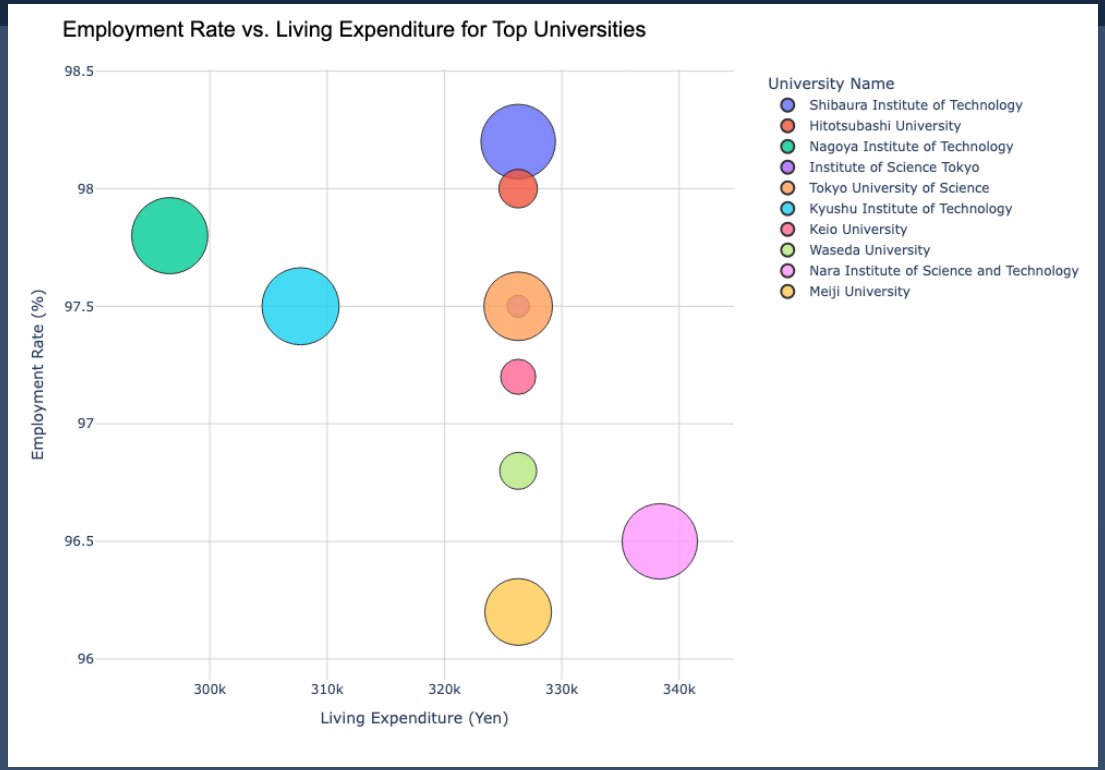
Most Optimal: Osaka University

→ Top-ranked universities outside Tokyo are cheaper to live in than Tokyo-based ones.

→ Tokyo universities consistently have higher living costs, regardless of rank.

→ High academic quality does not require high living expenditure.

Highest Employment Rate VS Their Living Expenditure



→ Employment rates remain high across universities, regardless of living expenditure.

→ Non-Tokyo universities offer comparable employability at lower living costs.

→ Higher living expenditure does not guarantee better employment outcomes.

Conclusions

1. Universities in **Tokyo** offers the **highest concentration** of high-ranked universities and high employment opportunities, although it comes with the **highest living expenditures**.

2. Universities in **Kyoto** and **Osaka** provide **similar academic quality** and **employability** to Tokyo, but they also have **lower living costs**, making them highly attractive.

3. **Public universities** all over Japan are the **most internationally welcoming**, with higher percentages of international students.

Overall – The best opportunities for international students are achieved by balancing internationalization, employability, academic performance, and living expenditure, rather than focusing only on rankings or major cities.